Math 521, Section 2
Midterm 1
Spring 2026

First name: _____ **Last name:** _____

Instructions:

- This exam contains 4 problems, and there are a total of 40 points available.
- Show all your work in the space provided. You may also use the backs of pages.
- No outside resources are allowed, including notes, calculators, textbooks, etc.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
Total:	40	

1. (10 points) Using induction, prove that for all $n \in \mathbb{N}$ we have

$$1 + 5 + 9 + \cdots + (4n - 3) = 2n^2 - n.$$

2. Let $f : A \to B$ be a function. **Prove or disprove** the following statements:
   (a) (5 points) For all $Y \subseteq B$, we have $f(f^{-1}(Y)) \subseteq Y$.
   (b) (5 points) For all $Y \subseteq B$, we have $f(f^{-1}(Y)) \supseteq Y$.

3. (10 points) Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded sets. Prove that if $a \leq b$ for all $a \in A$ and for all $b \in B$, then $\sup A \leq \inf B$.

4. (10 points) Prove that if $\{a_n\}_{n\geq 1}$ is a sequence of positive real numbers that converges to $a > 0$, then the sequence $\{\sqrt{a_n}\}_{n\geq 1}$ converges to $\sqrt{a}$.

Extra paper

1.) (Similar to HW1#1.)

Base case: For $n=1$, we have $1 = 2\cdot 1^2 - 1$, which is true.

Inductive step: Assume

$$1+5+\ldots+(4n-3) = 2n^2-n$$

for some $n \geq 1$. Adding $(4n+1)$ to both sides,

$$\begin{aligned} 1+5+\ldots+(4n-3)+(4n+1) &= 2n^2-n+(4n+1) \\ &= 2(n^2+2n+1)-n-1 \\ &= 2(n+1)^2-(n+1). \end{aligned}$$

So the formula holds for $n+1$.

By induction, we conclude the formula is true $\forall n \geq 1$. $\square$

2.) (Similar to HW1#3b)

(a.) <u>Prove</u>: Fix $Y \subseteq B$ and $y \in f(f^{-1}(Y))$. Then $\exists\, x \in f^{-1}(Y)$ s.t. $f(x) = y$. As $x \in f^{-1}(Y)$, then $f(x) \in Y$, and so $y = f(x) \in Y$.

(b.) <u>Disprove</u>: Consider $f: \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ and $Y = \{-1, 1\}$. Then

$$f^{-1}(\{-1,1\}) = \{-1,1\}, \qquad f(\{-1,1\}) = \{1\}$$

and so

$$f(f^{-1}(\{-1,1\})) = f(\{-1,1\}) = \{1\} \not\subseteq \{-1,1\}$$

(By HW1#3b, we just need to pick a function f that is not surjective.)

3.) (This is HW3#6a.)

As $A, B \subseteq \mathbb{R}$ are nonempty and bounded, $\sup A$, $\inf B \in \mathbb{R}$ exist.

Let $b \in B$. As $a \leq b$ $\forall a \in A$, then b is an upper bound for A, and so

$$\sup A \leq b.$$

This is true $\forall b \in B$. So $\sup A$ is a lower bound for B, and therefore

$$\sup A \leq \inf B. \qquad \square$$

4.) (This is HW4#5.)

Fix $\varepsilon > 0$. As $\lim_{n\to\infty} a_n = a$ and $a > 0$, then $\exists N \in \mathbb{N}$ s.t.

$$|a_n - a| < \sqrt{a} \cdot \varepsilon \qquad \forall n \geq N.$$

Then for $n \geq N$ we have

$$|\sqrt{a_n} - \sqrt{a}| = \left| \frac{(\sqrt{a_n} - \sqrt{a})(\sqrt{a_n} + \sqrt{a})}{\sqrt{a_n} + \sqrt{a}} \right| = \frac{|a_n - a|}{\sqrt{a_n} + \sqrt{a}}$$

$$\underset{\sqrt{a_n} > 0}{<} \frac{|a_n - a|}{\sqrt{a}} < \frac{\sqrt{a} \cdot \varepsilon}{\sqrt{a}} = \varepsilon$$

So $\lim_{n\to\infty} \sqrt{a_n} = \sqrt{a}$. $\square$

Math 521, Section 2
Practice Midterm 1
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2	10	
3	10	
4	10	
Total:	40	

1. (10 points) Let $(F, +, \cdot)$ be a field. Prove that if $a, b \in F \setminus \{0\}$, then $a^{-1} \cdot b^{-1} = (ab)^{-1}$.

   Make sure that each step follows from a property of fields that we stated in lecture. You do not need to cite each fact by name; just show your logic one step at a time.

2. (a) (3 points) State the definition of: the set A is *countable*.

   (b) (7 points) Prove that the set

$$B = \{(x, y) \in \mathbb{R}^2 : x \in \mathbb{N} \text{ and } y \in \{0, 1\}\}$$

   satisfies the definition in part (a).

3. (10 points) Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded sets, and define

$$S = \{a + b : a \in A \text{ and } b \in B\}.$$

Prove that $\sup S = \sup A + \sup B$.

4. Let $\{a_n\}_{n\geq 1}$ be a sequence and $a \in \mathbb{R}$. **Prove or disprove** the following statements:
   (a) (5 points) If $\{a_n\}_{n\geq 1}$ converges to a, then the sequence $\{|a_n|\}_{n\geq 1}$ converges to $|a|$.
   (b) (5 points) If the sequence $\{|a_n|\}_{n\geq 1}$ converges to $|a|$, then $\{a_n\}_{n\geq 1}$ converges to a.

Extra paper

1.) (This is HW3 # 1e.)

Fix $a, b \neq 0$. Then

$$\begin{aligned}(a^{-1}\cdot b^{-1})\cdot(ab) &= (a^{-1}\cdot b^{-1})\cdot(ba)\\ &= (a^{-1}\cdot(b^{-1}\cdot b))\cdot a\\ &= (a^{-1}\cdot 1)\cdot a\\ &= a^{-1}\cdot a\\ &= \underline{1}\end{aligned}$$

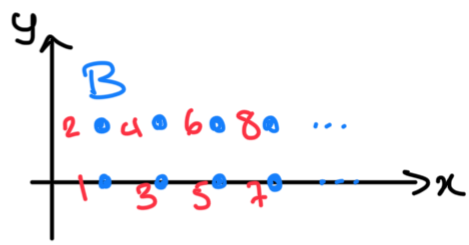
So $a^{-1}\cdot b^{-1}$ is the multiplicative inverse $(ab)^{-1}$ of ab. $\square$

2.) (This is a special case of HW2 # 1.)

(a.) $\exists\ f: A \to \mathbb{N}$ bijective

(b.) Define $f: B \to \mathbb{N}$ by

$$f(x,y) = \begin{cases} 2x-1 & \text{if } y=0 \\ 2x & \text{if } y=1 \end{cases}$$



Want: f bijective. Consider $g: \mathbb{N} \to B$,

$$g(n) = \begin{cases} \left(\frac{n+1}{2}, 0\right) & \text{if } n \text{ odd} \\ \left(\frac{n}{2}, 1\right) & \text{if } n \text{ even} \end{cases}$$

Then

$$(f\circ g)(n) = \begin{cases} f\left(\frac{n+1}{2}, 0\right) = 2\left(\frac{n+1}{2}\right)-1 = n & \text{if } n \text{ odd} \\ f\left(\frac{n}{2}, 1\right) = 2\left(\frac{n}{2}\right) = n & \text{if } n \text{ even} \end{cases}$$

$$(g\circ f)(x,y) = \begin{cases} g(2x-1) = \left(\frac{(2x-1)+1}{2}, 0\right) = (x,0) & \text{if } y=0 \end{cases}$$

$$\lfloor g(2x) = \left(\tfrac{2x}{2}, 1\right) = (x, 1) \qquad \text{if } y=1$$

for all $n \in \mathbb{N}$ and $(x,y) \in B$. So f is bijective by HW. $\square$

3.) (This is HW3 #5.)

As $A, B \subseteq \mathbb{R}$ are nonempty and bounded, $\sup A$, $\sup B \in \mathbb{R}$ exist.

① $\sup A + \sup B$ is an upper bound for S:

As $\sup A$, $\sup B$ are upper bounds for A, B,

$$a \leq \sup A \quad \forall a \in A, \qquad b \leq \sup B \quad \forall b \in B$$

$$\Rightarrow \quad a+b \leq \sup A + \sup B \quad \forall a \in A, \ \forall b \in B$$

② L upper bound for $S \Rightarrow L \geq \sup A + \sup B$:

Suppose not. Then L is an upper bound for S, but $L < \sup A + \sup B$. Set

$$\varepsilon = \tfrac{1}{2}(\sup A + \sup B - L) > 0.$$

As $\sup A - \varepsilon < \sup A$, then $\exists\, a_0 \in A$ s.t.

$$a_0 > \sup A - \varepsilon.$$

As $\sup B - \varepsilon < \sup B$, then $\exists\, b_0 \in B$ s.t.

$$b_0 > \sup B - \varepsilon.$$

Together,

$$a_0 + b_0 > \sup A + \sup B - 2\varepsilon = L.$$

This contradicts that L is an upper bound for S.

Collecting ① and ②, we conclude $\sup S$

$= \sup A + \sup B.$ $\square$

4.) (This is HW4 #1.)

(a.) <u>Prove</u>: Fix $\varepsilon > 0$. As $\lim_{n\to\infty} a_n = a$, $\exists N \in \mathbb{N}$ s.t.

$$|a_n - a| < \varepsilon \quad \forall n \geq N$$

Then for $n \geq N$ we have

$$\big||a_n| - |a|\big| \leq |a_n - a| < \varepsilon$$

So $\lim_{n\to\infty} |a_n| = |a|$. $\square$

(b.) <u>Disprove</u>: Consider $a_n = (-1)^n$. Then:

- $\{|a_n|\}_{n\geq 1}$ converges: since $|a_n| = 1$ $\forall n \geq 1$.
- $\{a_n\}_{n\geq 1}$ diverges: by lecture $\square$